MOTOR INSURANCE MANAGEMENT DATABASE SYSTEM

A Database Systems Project Report

AHABWE CEDRIC

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Course: Database Systems

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1. Executive Summary

This report presents the development of a **Motor Insurance Management Database System** using MySQL. The system was designed to manage customers, vehicles, insurance policies, claims, and payments for a motor insurance company.

The system provides efficient data management, faster claims processing, policy tracking, and reporting capabilities.

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2. Business Case

a. <u>Application Domain of the System</u>

The Motor Insurance Management System supports the core operations of a motor insurance company. Key processes include customer registration, vehicle details recording, policy issuance and renewal, claims management, and premium payments.

Main user groups served:

- Insurance Agents / Underwriters
- Claims Adjusters
- Customers (policyholders)
- Finance / Accounts Department
- Management (for reports and decision making)

The system automates manual processes that are currently done using spreadsheets and paper files, reducing errors and improving service delivery.

b. <u>Analysis of Benefits</u>

Tangible Benefits:

- Faster policy issuance and renewal
- Reduced claims processing time from weeks to days
- Accurate premium calculations and reduced revenue leakage
- Better tracking of expiring policies leading to higher renewal rates

Intangible Benefits:

- Improved customer satisfaction and retention
- Better risk assessment and fraud detection
- Enhanced company reputation through efficient service
- Easier regulatory compliance and reporting

c. <u>Analysis of Costs</u>

Tangible Costs: Development cost is minimal (free MySQL).

Intangible Costs: Initial staff training and possible resistance to new system.

Cost-Benefit Conclusion: Benefits significantly outweigh costs. The system is economically feasible.

d. <u>User Issues</u>

Most users (especially agents and claims officers) should have basic computer skills. Though this system uses simple interfaces and can be learned within 1–2 days of training.

e. <u>Risks Assessment</u>

- **Technical Risk:** Data loss or system downtime → mitigated by regular backups.
- **Security Risk:** Leakage of customer and vehicle data → mitigated by user authentication and MySQL privileges.
- **Regulatory Risk:** Changes in insurance laws → system is flexible for updates.

f. <u>Limitations of the System</u>

This version of database does not support high-volume concurrent users, advanced analytics (AI fraud detection), mobile app integration perhaps it is only limited to motor insurance only (not life or health insurance).

Comparable Real-world System: Jubilee Insurance's motor insurance . With strong integration between policy, claims, and payment modules, and the value of real-time reporting for management decisions.

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3. Design

Logical Design

a<u>. List of Entities</u>

- Customer
- Vehicle
- Policy
- Claim
- Payment

b. <u>List of Entities and Attributes</u>

- **Customer**: CustomerID (PK), FirstName, LastName, DateOfBirth, Address, Phone, Email, NationalID, RegistrationDate
- **Vehicle**: VehicleID (PK), CustomerID (FK), PlateNumber, Make, Model, YearOfManufacture, ChassisNumber, Color, EngineCapacity
- **Policy**: PolicyID (PK), PolicyNumber, CustomerID (FK), VehicleID (FK), PolicyType, StartDate, EndDate, PremiumAmount, Status
- **Claim**: ClaimID (PK), ClaimNumber, PolicyID (FK), ClaimDate, AccidentDate, Description, ClaimAmount, Status
- **Payment**: PaymentID (PK), PolicyID (FK), Amount, PaymentDate, PaymentMethod, Status

c . <u>Entity Relationships</u>

- Customer **1:N** Vehicle (One customer can own many vehicles)
- Customer **1:N** Policy (One customer can have many policies)
- Vehicle **1:1** Policy (One vehicle has one active policy at a time)
- Policy **1:N** Claim (One policy can have multiple claims)
- Policy **1:N** Payment (One policy can have multiple payments)

Physical Design

The physical design is well displayed in the slides executed on the motor_insurance database.

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4. Implementation

The database was successfully implemented in MySQL.

Database Name: motor_insurance

Tables Created:

- Customer
- Vehicle
- Policy
- Claim
- Payment